

Nadir ZOUAOUI

Electrical Engineer - Energy Efficiency, Renewables & 3D Simulation / Digital Twins

Oran, Algeria | nadir.zouaoui@hotmail.com | +213 656 060 214 | www.linkedin.com/in/nadir-zouaoui/

Portfolio: nadirzouaoui.github.io | behance.net/nad04799c26

Class B driver's license | Personal vehicle | National service: exempt



PROFESSIONAL SUMMARY

State engineer in electrical engineering with four years of experience across renewable energy, Oil & Gas and industrial systems. Delivered over 1 MWp of solar capacity, wrote energy sobriety plans for industrial clients, and audited 66 remote SCADA stations down to the condition of each asset. In parallel, I build real-time 3D simulations (Blender, Godot 4) used as interactive training modules, the same toolset behind operational digital twins. I want to put both skill sets to work on measuring and reducing the carbon footprint of industrial assets.

CORE SKILLS

Energy sobriety and efficiency plans | Photovoltaic design, sizing and simulation | Load and power quality analysis | Asset condition audits and SCADA data | Hazardous areas (ATEX) | Commissioning | 3D modelling and real-time simulation | Interactive digital twins and training modules

TECHNICAL TOOLS

Archelios PV simulation | Caneco BT | EPLAN Electric P8 + Pro Panel | AutoCAD Electrical | TIA Portal, STEP 7 | Siemens S7-1200 | Blender 3D | Godot 4 (GDScript, WebGL2 export) | SCORM 1.2 | Adobe Suite | Microsoft Excel

CERTIFICATIONS

Low Voltage Design Fundamentals and Caneco BT Training - Lokelect Engineering

September 2026 | 40 hours | Online | Certificate ID 2b3376a9aea18854

- LV design: earthing systems, power balance, cable sizing, voltage drop, selectivity basics.
- Caneco BT: project setup, modeling, results analysis and design iteration.

Electrical Equipment in Hazardous Areas (EEHA) - ROAR Training Solutions (Australian RTO)

2026 | Online | Australian national competency units (AS/NZS 60079, aligned with IEC 60079)

- Classified hazardous areas, explosion-protection requirements and planning of Ex installations (UEEHA0004, UEEHA0022, UEEHA0029).
- Installation of explosion-protected equipment and wiring systems (UEEHA0025).
- Detailed, visual and close inspection of hazardous-area installations (UEEHA0020, UEEHA0038).
- Maintenance of Ex equipment and periodic inspection and maintenance programs (UEEHA0026, UEEHA0023, UEECD0011).

PROFESSIONAL EXPERIENCE

Photovoltaic Design Engineer - Sungy SPA

October 2022 - July 2023 | Algeria

- Formulated energy sobriety plans and solutions for clients, identifying consumption reductions before sizing solar.
- 3D layout, sizing and simulation of PV systems; delivered over 300 kWp of installed solar capacity.
- Electrical sizing and drafting of electrical diagrams for solar installations.
- Commissioning and monitoring of solar installations; design of deployable mobile solar solutions.

Electrical Engineer - RMBTECH EURL

March 2025 - Present | Algeria

- SCADA gas pipeline projects (Sonatrach): site audit of 66 isolated solar/grid-powered stations on the GK3 pipeline, producing an equipment condition status for every station (good, damaged, vandalized); deployment on GK1/GK2 and pre-engineering of GR1/2/4/6 (integrator: SUPCON).
- Critical equipment and ATEX: engineering for the replacement of an 8500 A rectifier for the AOA electrolysis system; design of an ATEX Ex "p" pressurized cabinet for Sonatrach GP2Z (Arzew).
- Industrial rehabilitation: electrical redesign (EPLAN), rewiring and automated commissioning (Siemens S7-1200) of a BOSSAR packaging machine for GBFoods (Jumbo).
- Agro-industry: electrical design for the SCADA system of the OAIC grain silo in Mesra.

Lead Electrical Engineer - PROSKID EURL

July 2023 - March 2025 | Algeria

- Designed and sized the off-grid solar power supplies feeding Oil & Gas chemical injection skids (clients ENGCB/GCB, SARPI).
- Commissioned 27 Milton Roy anticorrosion dosing skids and their solar installations.
- Designed and built an ATEX Ex "d" flameproof enclosure for a corrosion inhibitor skid.
- Power supply, power and control schematics in EPLAN Electric P8 and AutoCAD Electrical.

Document Controller - CEVIELEC SPA (Cevital Electrique)

March 2022 - October 2022 | Algeria

- ALRAR Gas Processing Complex re-instrumentation: document revisions and markups, planning, cabinet wiring, instrument installation, loop checks and troubleshooting.

FREELANCE

3D Training Simulations - ROAR Training Solutions (Australian RTO)

- Developed first-person 3D playable exercises for electrical competency training: a 3D substation switching exercise and a Low Voltage Rescue and CPR simulation.
- Modelled equipment and environments in Blender; built interaction logic, procedure checklists and scoring in Godot 4.
- Packaged as SCORM 1.2 modules running in the browser (WebGL2) for delivery through the client's LMS.

5.4 MW Industrial Solar Installation - SARL Ibrahim & Fils (IFRI), Akbou, Bejaia

- Rooftop PV design for a zero-injection self-consumption plant over several production sites: JinkoSolar 635 Wc bifacial modules, Huawei SUN2000-150K inverters, 10 x 241 kWh battery storage.
- Load and power quality analysis; string and inverter zoning; DC/AC cable sizing optimized for voltage drop and cost.

3D Visualization & Motion Design

- Engineering 3D animation, product renders, exhibition stand visualization and motion design (portfolio: behanee.net/nad04799c26).

EDUCATION

State Engineer + Master II, Electrical Engineering

Ecole Nationale Polytechnique d'Oran - Maurice Audin (ENPO-MA), Oran, Algeria | 2016 - 2021

- Valedictorian, class of 2018-2021.
- Master's thesis: design, build and commissioning of a 5 kW autonomous PV microgrid (3 kW PV, 800 Ah battery bank, 50 kVA genset, three-source hybrid inverter).

LANGUAGES

Arabic - Native | French - Bilingual, C1 certified (DALF 76/100) | English - Fluent